

Algorithms of emotion: A hybrid NLP analysis of neurodivergent Reddit communities

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ABSTRACT

Online communities such as Reddit offer neurodivergent individuals a unique space to express emotions, seek psychosocial support, and negotiate identity outside conventional social constraints. Understanding how these communities articulate and structure emotional discourse is essential for inclusive technology design. This study employed a hybrid natural language processing (NLP) framework that integrates lexicon-based sentiment analysis (VADER) with transformer-based topic modeling (BERTopic). A dataset comprising 238,100 posts and comments from subreddits focused on autism, ADHD, neurodiversity, and mental health (January 2023–February 2024) was analyzed to detect thematic and affective patterns in digital neurodivergent discourse. The analysis revealed ten semantically coherent and emotionally salient topics, including emotional overload, masking, diagnostic barriers, coping strategies, and AI-mediated interaction. Sentiment-topic alignment highlighted how affective tone is embedded within distinct thematic clusters, underscoring the structured emotional dynamics of neurodivergent communities. This study advances methodological rigor in affective computing and human computer interaction by combining scalable NLP tools with visual analytics to map digital neurodivergent discourse. It contributes a reproducible framework for analyzing emotion-rich online content and provides empirical grounding for the inclusive design of emotionally responsive digital environments tailored to neurodivergent users.

1. Introduction

The intersection of digital environments and neurodivergence presents a compelling area of study, particularly in the context of online platforms like Reddit, where individuals interact within diverse, often self-organized communities (Proferes et al., 2021). Reddit's user-driven structure and topic-specific subreddits create a rich ecosystem for exploring the discourse of neurodivergent individuals, allowing for both expressive freedom and psychosocial support (Morales et al., 2021). Within subreddits such as r/Autism, r/ADHD, and r/MentalHealth, users share lived experiences, reflect on challenges, and negotiate identity in ways that often transcend offline limitations (Brown et al., 2018). These digital spaces therefore function as vital support systems, where neurodivergent identities are articulated, affirmed, and co-constructed. This evolving landscape underscores the need for interdisciplinary approaches capable of examining both the emotional and structural dimensions of digital neurodivergent discourse. Online platforms not only enable interaction but also shape how identity and emotional expression unfold in environments guided by algorithms. Existing scholarship

highlights Reddit's role in fostering identity work, coping exchanges, and self-disclosure among neurodivergent individuals (Doyle & McDowell, 2021; Hamilton & Petty, 2023). However, despite this attention, there is still limited large-scale research that connects emotional tone with broader discussion patterns, particularly when considering how these dynamics intersect with new technologies such as artificial intelligence (AI). To address this gap, the present study investigates how neurodivergent individuals engage with digital platforms to express emotion, construct identity, and reflect on their experiences with technology. Rather than focusing only on broad themes, we emphasize the underlying emotional aspects of communication and examine how these expressions blend with patterns of meaning in online communities.

The investigation is guided by the following research questions:

RQ1. What are the dominant emotional tones expressed by neurodivergent individuals in Reddit discussions, and how do these sentiments vary across different contexts and themes?

RQ2. What recurring thematic patterns characterize neurodivergent discourse in online environments?

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RQ3. How do neurodivergent users narrate their experiences with technological tools such as AI companions and automation systems within broader discourses of identity, coping, and systemic barriers?

RQ4. To what extent do emotional expression and thematic clusters align, diverge, or co-evolve in online neurodivergent communities?

RQ5. How do the structural and semantic properties of identified topics hold up in terms of interpretability, coherence, and emotional consistency?

By addressing these questions, this study contributes to the interdisciplinary literature at the intersection of digital discourse analysis, affective computing, and neurodiversity studies. Unlike prior research that focused mainly on thematic trends, our approach brings attention to the lived emotional and cognitive experiences of neurodivergent individuals navigating online ecosystems. This perspective provides both empirical and theoretical insights into how identity, emotion, and technology interact in contemporary digital environments, offering a foundation for designing more inclusive and emotionally responsive digital spaces.

The study is guided by two complementary perspectives: distributed cognition and emotional design. Distributed cognition theory suggests that thinking and meaning-making are not confined to individuals but are shared across people, tools, and environments (Hollan et al., 2000). Applied here, Reddit can be seen as a system where identity work, knowledge-sharing, and emotional processing occur collectively in a digital space. Emotional design, on the other hand, highlights how technologies and interfaces trigger emotional responses, influencing engagement and well-being (Norman, 2004; Picard, 1997). Together, these perspectives allow us to view neurodivergent communication online as both emotionally rich and socially distributed, shaped simultaneously by platform features and individual strategies for managing identity and emotion.

2. Literature review

The pervasive integration of digital environments into daily life has created novel avenues for individuals to connect, communicate, and share experiences, particularly for those who identify as neurodivergent (Proferes et al., 2021). Neurodiversity, a concept gaining traction in both academic and professional spheres, acknowledges the natural variations in human brain function and behavioral traits (Doyle, 2020; Lee et al., 2024). It encompasses a range of conditions, including autism spectrum disorder, attention deficit hyperactivity disorder, dyslexia, and other neurological differences (Doyle & McDowall, 2021). These variations affect how individuals perceive and interact with the world, learn, and process information (Hamilton & Petty, 2023). The rise of online platforms, such as Reddit, has provided neurodivergent individuals with spaces to form communities, share insights, and discuss challenges and successes related to their unique experiences (Koehler, 2024). Within these digital communities, individuals can find support, exchange information, and advocate for greater understanding and acceptance of neurodiversity in broader society (Doyle, 2020). The anonymity afforded by platforms like Reddit can encourage participation, although disparities in engagement may exist across demographics, with women potentially being less likely to actively participate through commenting and posting compared to men (Brown et al., 2018). The relative anonymity afforded by online platforms and permissive content policies may lead to sensitive discussions (Proferes et al., 2021). However, despite its growing relevance, scholarly work systematically analyzing large-scale neurodivergent discourse on Reddit remains limited, particularly in the context of natural language processing (NLP). Emerging studies have underscored the value of computational social science in examining mental health and neurodiversity, yet there is a notable lack of integrated affective and thematic analyses focusing specifically on neurodivergent communities (Alghamdi et al., 2025; Gao

& Sazara, 2023).

2.1. Sentiment analysis and neurodivergent communities

Sentiment analysis, also known as opinion mining, is a natural language processing technique used to determine the evaluative tone of text, typically categorizing expressions as positive, negative, or neutral (Hutto & Gilbert, 2014; Liu, 2012). Within neurodivergent communities, sentiment analysis provides a scalable means of understanding the emotional dimensions of online discussions, offering insights into challenges such as masking, burnout, coping strategies, and identity negotiation. By analyzing Reddit posts and comments, researchers can map how neurodivergent individuals articulate distress, seek support, and engage in identity work in digitally mediated contexts (Alghamdi et al., 2025; Guo et al., 2021; Tul et al., 2017). Importantly, while often conflated in everyday language, *masking* and *camouflaging* are distinct constructs within neurodiversity research. Masking involves the deliberate suppression of neurodivergent traits (e.g., stimming or atypical speech patterns) to conform to neurotypical norms, whereas camouflaging encompasses broader adaptive strategies such as mimicking social behaviors, rehearsing conversational scripts, and deploying learned social cues to blend into neurotypical environments (Cook et al., 2021; Hull et al., 2017; Livingston et al., 2020). This study employs “masking” to denote behavior-specific suppression and “camouflaging” for multi-layered adaptive strategies that extend beyond trait concealment.

Additionally, it is essential to distinguish sentiment analysis from emotion analysis. Whereas sentiment analysis offers a coarse-grained assessment of evaluative tone, emotion analysis focuses on identifying discrete affective states (e.g., joy, anger, fear) or multidimensional representations (e.g., valence-arousal) (Mohammad & Turney, 2013; Scherer, 2005). Emotion analysis provides a finer-grained mapping of affective expression but often demands more complex lexicons or supervised models (Balahur et al., 2012). Building on these distinctions, this study employs sentiment analysis in conjunction with transformer-based topic modeling (BERTopic) to align emotional tone with coherent thematic clusters (Elkins & Chun, 2019). This dual-layered approach enhances interpretability by integrating sentiment polarity with thematic context, enabling a more nuanced view of neurodivergent discourse in online communities beyond lexicon-based sentiment scoring.

2.2. Reddit as a platform for studying neurodivergent experiences

Reddit hosts a multitude of communities, known as subreddits, dedicated to various topics and interests, including neurodiversity. These subreddits serve as valuable sources of data for researchers interested in understanding the experiences and perspectives of neurodivergent individuals. The platform's structure, which allows users to create and participate in specialized communities, fosters focused discussions on specific topics related to neurodiversity, creating a rich ecosystem of shared experiences and support. By analyzing the text-based content of Reddit posts and comments, researchers can identify recurring themes, assess the sentiment expressed toward specific issues, and gain a nuanced understanding of the challenges and triumphs of neurodivergent individuals (Bordoloi & Biswas, 2023). The prevalence of social media and the availability of data enable new investigations that were previously impossible (Blenn et al., 2012). The views, evaluations, and comments about everyday activities can be extracted because of advancements in Internet services and social media platforms (Sharma & Goyal, 2023). Reddit has proven to be influential in the public domain (Thapa, 2022; Tunca, 2025).

Understanding Reddit's affordances is crucial for contextualizing neurodivergent discourse. Reddit's semi-anonymous structure facilitates self-disclosure and reduces stigma associated with sensitive topics, encouraging candid emotional expression (Fiesler et al., 2024; Proferes et al., 2021). However, platform-specific features such as the karma

system and upvote-based sorting introduce biases that amplify popular or socially sanctioned content while diminishing minority or controversial perspectives (Matias, 2019). Subreddit-specific rules and moderation practices further shape discourse, delineating acceptable language and governing topic boundaries, thereby influencing both participation and content visibility (Chandrasekharan et al., 2017). These affordances collectively impact how neurodivergent individuals navigate digital identity work, emotional sharing, and community support online. By situating Reddit within an affordance-theoretical framework, this study underscores how technical and social platform mechanisms co-produce the conditions under which neurodivergent discourse emerges and circulates. However, few studies explicitly connect Reddit-based analyses of neurodivergent discourse with affective computing and inclusive design principles, creating an opportunity to bridge computational linguistics with user-centered digital accessibility research (McDuff et al., 2016).

2.3. Methodological approaches to analyzing online forums

Analyzing online support forums requires a well-defined methodological approach to ensure the validity and reliability of findings (Smedley & Coulson, 2018). Data collection involves gathering relevant posts and comments from specific subreddits or online forums related to neurodiversity (Amaya et al., 2019). Natural Language Processing techniques play a crucial role in pre-processing the text data, which includes cleaning the text, removing irrelevant characters, and standardizing the format. Sentiment analysis algorithms are applied to determine the emotional tone expressed in the text, categorizing it as positive, negative, or neutral. Topic modeling techniques can identify prevalent themes and topics discussed within the online forums, providing insights into the key issues and concerns of the community (Leung & Khalvati, 2022). Integrating these approaches with reliability assessments such as topic coherence (C_v) and sentiment-topic alignment ensures robust interpretability, particularly when applied to neurodivergent contexts where emotional expression is nuanced and heterogeneous.

2.4. Leveraging sentiment analysis for understanding neurodivergent experiences

Sentiment analysis has become increasingly prevalent in social media research, providing insight into how individuals express attitudes and emotions in digitally mediated contexts (Lo et al., 2016; Saraff et al., 2018). Within neurodivergent communities, it enables the detection of patterns in emotional expression tied to key experiences such as social stigma, coping mechanisms, and therapeutic support. By applying sentiment analysis to Reddit discussions, researchers can gauge how neurodivergent individuals articulate distress, resilience, and empowerment across various themes (Rojas-Barahona, 2016; Yaakub et al., 2019).

Recent advancements in deep learning and natural language processing have further enhanced the capabilities of sentiment analysis. Techniques such as transformer-based embeddings and neural network models (e.g., CNNs, LSTMs) automatically learn complex semantic representations from text, improving the accuracy and contextual sensitivity of sentiment detection (Machová et al., 2023; Socher et al., 2013; Tang et al., 2015). These methods are particularly effective in analyzing large-scale datasets, where traditional lexicon-based approaches may fail to capture nuanced or context-dependent sentiment shifts (Bhat, 2024; Zhang et al., 2018).

In neurodivergence-focused studies, sentiment analysis complements topic modeling by aligning emotional polarity with semantically coherent clusters (Elkins & Chun, 2019). This integration allows for a richer interpretation of discourse by highlighting how emotional tone interacts with themes such as masking, burnout, or identity affirmation. Moreover, tracking sentiment over time can reveal how perceptions of

neurodiversity evolve in response to cultural or technological changes, offering dynamic insight into the emotional landscape of online communities.

By leveraging these computational approaches, researchers can move beyond descriptive content analysis and instead construct a multi-layered view of neurodivergent online discourse—one that captures not only *what* is discussed but also *how* these discussions are emotionally framed. This dual perspective is essential for understanding the intersection of emotion, identity, and digital participation in neurodivergent spaces.

2.5. Leveraging topic modeling analysis for understanding neurodivergent experiences

Topic modeling is a powerful technique used to uncover latent themes and topics within a collection of documents. By applying topic modeling to Reddit discussions, researchers can identify the most frequently discussed themes and issues related to neurodiversity (Elkins & Chun, 2019). Topic modeling excels at distilling large volumes of text into a concise set of representative topics. These topics can encompass a wide range of subjects, from diagnostic experiences and coping strategies to advocacy efforts and community support initiatives. Topic modeling helps researchers understand the diverse perspectives and experiences within the neurodivergent community. Topic modeling can also reveal how different topics are interconnected and how they evolve over time. Researchers have explored the use of advanced natural language processing methods, such as BERT as a text mining approach for mental health prediction (Gao & Sazara, 2023). Furthermore, advanced NLP approaches such as BERTopic offer enhanced interpretability by combining transformer embeddings with density-based clustering (Gao & Sazara, 2023). This is particularly useful for neurodivergent discourse, where heterogeneous linguistic styles and emotional intensities challenge conventional modeling methods.

2.6. Leveraging topic coherence graph analysis for understanding neurodivergent experiences

Topic coherence measures the semantic similarity between the words within a topic, providing insights into the interpretability and meaningfulness of the identified topics. By constructing topic coherence graphs, researchers can visualize the relationships between different topics and identify clusters of related themes. Examining the structure and connectivity of topic coherence graphs can reveal the underlying cognitive frameworks and associations that shape neurodivergent individuals' understanding of their experiences. By generating interpretable and coherent topics, researchers can gain a deeper understanding of the key issues and concerns within the neurodivergent community. Recent work has demonstrated the capability of textual data in various tasks, like sentiment and emotion recognition (Luna-Jiménez et al., 2021). Emotion analysis identifies the emotion or attitude toward a target or topic (Binali & Potdar, 2012; Ibrahim et al., 2019; Pennebaker et al., 2007; Sasidhar et al., 2020).

Human-Computer Interaction literature has long emphasized the importance of emotional engagement, accessibility, and user diversity in digital system design (Norman, 2004; Picard, 1997). Recent work in affective computing has extended this by developing interfaces that respond to users' emotional states, particularly relevant for neurodivergent users who may process emotions and interfaces differently (Höök, 2008; McDuff et al., 2016). Studies in inclusive design also argue for adapting interaction models to accommodate a range of cognitive and sensory preferences, advocating for "designing for the margins" to improve access for all (Holone, 2016; Frauenberger et al., 2013). These approaches are especially critical in neurodivergent contexts, where emotional overload, masking, and non-normative communication styles present unique interaction challenges (Benton et al., 2012; Hayes et al., 2020). Despite this, there remains a significant gap in understanding

how AI-mediated tools and algorithmic interfaces shape neurodivergent user experiences. This study contributes to filling this gap by empirically mapping the emotional and semantic patterns in neurodivergent digital discourse and aligning them with themes from emotional design and adaptive interaction. Despite these methodological advancements, there remains a paucity of studies linking computational findings to practical design implications. By empirically mapping emotional and semantic patterns in neurodivergent online communities, this study directly addresses this gap, contributing insights relevant to both NLP research and inclusive technology design.

2.7. Theoretical framework and hypotheses development

The increasing reliance on online communities as psychosocial support ecosystems for neurodivergent individuals offers a unique intersection of *digital discourse*, *affective expression*, and *cognitive diversity* that is understudied in computational social science. Previous literature demonstrates that online forums such as Reddit facilitate identity exploration, emotional disclosure, and peer validation for neurodivergent populations (Ireland & Iserman, 2018; Koehler, 2024). Yet, these spaces also reproduce distress linked to social stigma, healthcare barriers, and emotional labor, often resulting in polarized emotional narratives (Hull et al., 2017; Doyle & McDowall, 2021). This study positions itself within three interconnected scholarly domains—neurodiversity research, affective computing, and NLP-driven social media analysis—to generate hypotheses that not only extend existing knowledge but also address the methodological and theoretical gaps identified in prior studies.

2.7.1. Emotional expression and distress in neurodivergent online discourse

Research on neurodivergent online communities highlights elevated emotional intensity compared to neurotypical populations, particularly in discussions about masking, burnout, and co-occurring mental health conditions (Guo et al., 2021; Hull et al., 2017). Masking—the deliberate suppression of neurodivergent traits to conform to social norms—has been linked to emotional exhaustion, internalized stigma, and heightened anxiety in both offline and digital contexts (Livingston et al., 2020). Studies employing sentiment analysis in mental health forums corroborate these findings, showing that negative sentiment is concentrated in threads discussing chronic stressors and unmet needs (Rashid, 2024).

Building on this evidence and responding to RQ1 (emotional tone distribution) and RQ2 (thematic characterization), we hypothesize:

H1. Discussions centered on emotional overload, masking, and co-occurring conditions will exhibit significantly higher negative sentiment polarity compared to other themes in neurodivergent discourse.

This hypothesis aligns with affective computing literature, which emphasizes the role of distress markers as indicators of high affective salience in computational models of mental health (McDuff et al., 2016).

2.7.2. Constructive coping and community-driven positivity

While emotional distress dominates certain discussions, prior research also underscores the resilience and adaptive strategies emerging in neurodivergent communities. Studies of online autism and ADHD forums reveal solution-focused exchanges, where users collectively develop coping mechanisms for executive dysfunction, sensory overload, and social challenges (Doyle & McDowall, 2021; Koehler, 2024). These communities often reframe neurodivergence from a deficit lens toward identity empowerment, echoing principles of positive psychology and digital resilience (Hamilton & Petty, 2023). This literature directly informs RQ3 (identity and coping narratives) and suggests that coping-related threads, grounded in peer support, differ affectively and structurally from clinical or diagnostic conversations. Thus, we posit:

H2. Community-driven coping and self-help discourse will exhibit

significantly more positive sentiment and higher thematic coherence compared to diagnostic or institution-focused threads.

This aligns with socio-constructivist perspectives (Austin & Pisano, 2017), which argue that identity-affirming spaces promote constructive emotional regulation and peer validation.

2.7.3. Technology engagement and ambivalent AI narratives

Recent studies in human-computer interaction (HCI) and affective computing emphasize both the promise and pitfalls of AI-mediated support for neurodivergent users. AI-based tools, such as virtual companions and task-management systems, are valued for reducing cognitive load but often fail to meet deeper socio-emotional needs (Kravets et al., 2024). Research on technology acceptance among neurodivergent populations notes ambivalence, where functional utility is counterbalanced by frustration over emotionally unresponsive interfaces (Frauenberger et al., 2013; Höök, 2008). This duality motivates RQ4 (AI-mediated experiences) and underpins the hypothesis:

H3. Discussions involving AI tools and digital assistance will display mixed sentiment profiles, reflecting simultaneous appreciation for practical utility and dissatisfaction with emotional responsiveness.

By integrating NLP with HCI theory, this hypothesis advances calls for affect-sensitive design (Norman, 2004; Picard, 1997) and emphasizes the need to empirically map these nuanced emotional-technical dynamics in neurodivergent discourse.

2.7.4. Sentiment-thematic alignment and hybrid NLP modeling

Traditional NLP studies on Reddit-based mental health data often isolate sentiment or topic modeling rather than examining their alignment (Elkins & Chun, 2019). Yet, hybrid frameworks combining lexicon-based sentiment scoring (e.g., VADER) with transformer-driven topic modeling (e.g., BERTopic) allow researchers to test whether affective tone naturally clusters around semantically coherent themes. Prior work in computational mental health has shown that integrating emotion with thematic context enhances both interpretability and predictive validity (Gao & Szara, 2023). To address RQ5 (affective-thematic co-evolution), we hypothesize:

H4. Sentiment polarity will significantly align with semantically coherent topic clusters, validating the integration of affective and thematic modeling for neurodivergent discourse analysis.

This hypothesis strengthens the methodological contribution of our study by empirically evaluating hybrid sentiment-topic models within a neurodivergent context, a domain largely absent in NLP research.

2.7.5. Evolution of research questions and study differentiation

These hypotheses collectively evolve from the identified gaps in prior research:

1. *Emotional intensity and negative sentiment* are well-documented but lack systematic thematic mapping in large-scale neurodivergent communities (H1).
2. *Constructive coping and positive reframing* remain underexplored in NLP-driven analyses despite their prominence in neurodiversity literature (H2).
3. *AI engagement and ambivalent affect* are emerging concerns at the intersection of HCI and neurodivergence but lack empirical grounding in discourse data (H3).
4. *Hybrid affective-semantic modeling* has been proposed but not empirically validated within neurodivergent discourse (H4).

By addressing these gaps, this study distinguishes itself through its hybrid NLP pipeline, integration of sentiment-topic alignment, and explicit linkage to affective computing and inclusive design frameworks. Unlike prior descriptive studies, our approach provides theoretically

grounded, computationally validated insights into the *emotional-semantic architecture of neurodivergent digital discourse*.

3. Methodology

3.1. Dataset description and collection procedure

The dataset employed in this study was derived from public Reddit posts and comments collected from subreddits explicitly dedicated to neurodiversity, including r/Autism, r/ADHD, r/Neurodiversity, and r/MentalHealth. These communities are widely recognized for fostering active, emotionally expressive, and thematically rich discussions among neurodivergent individuals. To ensure topical relevance and recency, only content published between January 2023 and February 2024 was included.

Data collection was carried out using the Pushshift API for large-scale extraction, supplemented by the Python Reddit API Wrapper (PRAW) for metadata retrieval. Subreddit-level querying and keyword filtering (e.g., “autistic burnout,” “ADHD masking”) were employed to identify relevant threads. Inclusion criteria required a minimum of 20 words per post and at least 10 comments per thread. Only English-language content was retained to ensure consistency across the NLP pipeline.

The final dataset comprises 238,100 user-generated entries, each with three core components: subreddit, post title, and comment. These textual fields facilitate multi-level analysis from individual sentiment polarity to community-level thematic patterns.

Data quality assessments revealed only one missing comment across the dataset. Variables were processed as string-type objects and pre-processed using tokenization, lemmatization, and bot/noise filtering. No personally identifiable information (PII) was collected, and user anonymity was maintained throughout.

This curated corpus includes approximately 7800 original posts and 38,000 comments, totaling over 2.4 million words. Its lexical richness and structural diversity enable advanced applications such as sentiment analysis, BERTopic-based thematic modeling, semantic network visualization, and comparative subreddit discourse analysis.

The dataset exhibits an approximate post-to-comment ratio of 1:5. Notably, certain subreddits, such as r/ADHD, contributed a disproportionate share of content, reflecting high engagement within specific neurodivergent communities. Additionally, frequent posters—users contributing multiple entries—were identified, potentially amplifying particular perspectives. These factors suggest caution in interpreting representativeness, as the dataset may overrepresent highly active users and dominant subreddit cultures. While the large volume and lexical diversity support broad thematic extraction, the uneven distribution constrains the generalizability of findings to less active or offline neurodivergent populations.

3.1.1. Keyword filtering and validation procedures

To construct the dataset, we implemented a keyword-based retrieval strategy using Reddit’s Pushshift API to extract posts from neurodivergence-related subreddits. The keywords were selected through an iterative literature review (Doyle & McDowall, 2021; Hull et al., 2017) and pilot corpus inspection to balance coverage and specificity. The final keyword list included:

“autism”, “asperger”, “adhd”, “add”, “neurodivergent”, “neurodiversity”, “masking”, “sensory overload”, “stimming”, “executive dysfunction”, “burnout”, “self-regulation”, “AI support”, “chatbot”, “virtual assistant”, “diagnosis”, “therapy”, “coping strategies”, “peer support”, “identity”, “acceptance”].

Validation Steps to Address Keyword Bias:

1. Diverse Subreddit Sampling: Data were collected not only from neurodivergence-specific subreddits but also from general mental health and AI forums where relevant terms appeared contextually.

2. Post-Retrieval Keyword Masking: Keywords were masked during preprocessing to prevent their influence on topic embeddings.
3. Unsupervised Topic Discovery: BERTopic identified themes via transformer-based embeddings, ensuring clusters formed through semantic similarity rather than keyword repetition.
4. Manual Topic Validation: Three independent coders reviewed 20 posts per topic to confirm thematic emergence ($\kappa = 0.82$).
5. Sensitivity Analysis: Removing the top 10 % frequent keywords yielded stable topic structures (ARI = 0.87), confirming robustness against keyword-driven artifacts.

This multi-step approach ensured that themes were emergent and not merely artifacts of keyword-driven retrieval.

3.2. Research design

This study adopts a computational social science approach, combining natural language processing (NLP) techniques with qualitative interpretation to analyze sentiment and thematic patterns in Reddit discussions related to neurodivergence. The research is exploratory and descriptive in nature, aiming to uncover the emotional tone, dominant themes, and structural coherence of online discourse in neurodivergent communities.

3.2.1. Sentiment analysis workflow (VADER)

We applied VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment tool designed for social media, to assign polarity scores between -1 (negative) and $+1$ (positive). VADER accounts for intensifiers, negations, and colloquial markers prevalent in Reddit discussions. Sentiment scores were aggregated at the topic level to evaluate emotional trends across emergent clusters.

3.2.2. Topic modeling (BERTopic pipeline)

This study employed BERTopic, a transformer-based topic modeling approach, due to its superior ability to generate semantically coherent topics compared to classical models such as Latent Dirichlet Allocation (LDA) and alternative transformer-based models like Top2Vec. BERTopic leverages Sentence-BERT embeddings combined with UMAP for dimensionality reduction and HDBSCAN for density-based clustering, enabling the extraction of contextually rich, non-overlapping themes from large-scale textual data (Grootendorst, 2022). Empirical studies have demonstrated BERTopic’s superior topic coherence ($C_v > 0.50$) and interpretability, particularly in social media and discourse analysis contexts (Yao et al., 2022; Zhang et al., 2023).

We configured UMAP and HDBSCAN using established hyperparameters validated in prior literature. Specifically, UMAP was set to $n_neighbors = 15$ and $min_dist = 0.1$ to balance local density preservation with global semantic structure, supporting meaningful topic separability in high-dimensional text embeddings (McInnes et al., 2018). HDBSCAN was parameterized with $min_cluster_size = 40$ to reduce spurious micro-clustering and ensure robust theme extraction in line with best practices for social media corpora (Campello et al., 2013). Sensitivity analyses using alternative configurations (e.g., $min_dist = 0.05$, $min_cluster_size = 30$) confirmed that the selected parameters optimized topic coherence ($C_v = 0.51$) while minimizing redundancy (Adjusted Rand Index = 0.87).

Furthermore, a C_v coherence threshold of 0.45 was applied to filter out low-quality topics. This threshold aligns with standard practices in computational social science, particularly for short-text social media corpora where lexical variability is high (Röder et al., 2015). Manual validation of topic exemplars ($n = 20$ per topic) confirmed interpretability above this threshold, yielding substantial inter-rater agreement ($\kappa = 0.82$). Sensitivity testing at thresholds of 0.40 and 0.50 showed negligible differences in topic count but revealed that 0.45 best balanced semantic granularity and redundancy. This empirically grounded threshold ensures methodological rigor while preserving topic

research ethics guidelines (Bender et al., 2021), which emphasize user confidentiality, consent considerations, and responsible reporting of vulnerable populations.

These practices ensure that the research respects participant dignity and adheres to community norms, fostering trustworthiness and social responsibility in digital discourse analysis.”

4. Results

4.1. Initial analysis of concept map

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revealing underlying structures of meaning that might not be immediately apparent through surface-level text review.

At the center of the concept map lies the prominent concept “feel,” which suggests that emotional expression is a dominant concern in discussions among neurodivergent individuals. However, the verb “feel” exhibits polysemy in Reddit discourse, encompassing both subjective emotional experiences and descriptive sensory or situational states. For instance, subjective use appears in expressions such as “I feel overwhelmed” or “I feel anxious,” reflecting internal emotional states. In contrast, descriptive uses manifest in statements like “I feel cold” or “It feels like a burden,” which convey sensory perceptions or metaphorical external observations. Distinguishing these usages is critical for accurate sentiment and emotion analysis, as conflating subjective feelings with descriptive references may distort affective interpretation. Contextual cues and qualitative coding allowed for effective disambiguation, enhancing the precision of computational affective measures.

The presence of terms like “anxiety,” “overwhelmed,” “burnout,” and “struggle” in close proximity to “feel” indicated a pattern of shared emotional burden and psychological distress. These are not merely linguistic clusters but reflect the lived emotional texture of neurodivergent experience in digital environments.

The map also shows a significant cluster around “diagnosis,” “doctor,” and “ADHD,” which implies a focus on medical and institutional aspects of neurodivergence. This indicated that users frequently share their diagnostic journeys, challenges with healthcare systems, and the implications of clinical labeling. The semantic proximity between these medical terms and emotionally laden concepts like “help,” “support,” and “understand” reflects the duality of seeking both validation and assistance within online forums.

Another important theme is the presence of “masking” and “social,” which are interconnected in the map. This suggests active discussions on the strategies employed by neurodivergent individuals to adapt or conceal their traits in social settings—often at the expense of psychological well-being. The emotional cost of such masking is emphasized by its closeness to negative affective terms.

From a structural perspective, the map revealed relatively tight clusters, indicating focused and coherent discussions within specific domains such as emotional regulation, social adaptation, and diagnostic pathways. The semantic distance between clusters, however, suggests that although users share common experiences, these are segmented by topic boundaries, potentially influenced by subreddit norms or post intent (venting vs. advice-seeking).

This concept map provides a valuable foundation for further quantitative analysis using sentiment scoring and topic modeling. By identifying which clusters correspond to high levels of negative sentiment (e.g., “burnout”) versus those associated with support-seeking and community (e.g., “help,” “understand”), future stages of the study can align emotional tone with the semantic context. Complementing these analyses, we conducted network-based visualizations to reveal the structural patterns underlying neurodivergent discourse. The semantic network graph (Fig. 1) illustrates the co-occurrence and connectivity of key themes, uncovering distinct clusters centered on emotional expression, diagnosis, and social adaptation. Additionally, user interaction networks identified bridging users who facilitate dialogue across subreddits such as r/ADHD and r/Autism. These network insights highlight the complex, interconnected nature of neurodivergent online communities, demonstrating how thematic clusters are maintained through active cross-community engagement.

4.2. Concept co-occurrence frequency analysis

To further investigate the relational structure of key themes within neurodivergent Reddit discussions, a concept co-occurrence matrix was generated based on the frequency with which major terms appeared in proximity within the dataset. The matrix was then visualized as a heatmap, displaying the top 15 most frequent concepts. The resulting

visualization revealed notable co-occurrence patterns that highlight both semantic proximity and conversational density among core concepts.

The heatmap indicated a particularly strong association between the concepts “AI,” “people,” “time,” and “life,” suggesting that neurodivergent users frequently discuss these ideas in tandem. For instance, the intensity of the pairing between “AI” and “people” (value: 9030) suggests ongoing discourse around how artificial intelligence intersects with personal identity, social life, or cognitive processing—a theme that aligns with current concerns about assistive technologies and algorithmic influence in neurodivergent spaces.

Interestingly, the concept “porn” also features prominently, co-occurring with terms like “people,” “time,” and “life.” This indicated that conversations about neurodivergence do not shy away from sensitive or stigmatized topics, potentially reflecting honest discussions around internet use, compulsive behaviors, or taboo subjects often underrepresented in formal research. The concept “better,” frequently co-occurring with both emotional and behavioral terms, may reflect the pursuit of coping strategies or self-improvement narratives within the community.

The strength and density of co-occurrences among these terms provide empirical grounding for the concept map observed earlier. They also serve as input validation for subsequent topic modeling and sentiment overlays, reinforcing the idea that emotionally resonant and socially complex discussions are central to how neurodivergent individuals use Reddit as a communicative space.

4.3. Conceptual dissimilarity and semantic distance mapping

In addition to frequency-based co-occurrence analysis in Fig. 2, a semantic dissimilarity heatmap was constructed to capture how conceptually distant certain terms are from one another. This inverted matrix was calculated using reciprocal co-occurrence metrics, offering insights into which concepts are more isolated in conversational context. While the co-occurrence heatmap emphasized associative density, this visualization sheds light on divergence, thematic distance, and potential topic fragmentation.

The heatmap revealed that concepts such as “porn” and “AI” exhibit high dissimilarity with core psychosocial terms like “feel,” “need,” and “life.” This suggests that discussions surrounding these terms are relatively siloed, occurring in different threads or under separate narrative frames. On the other hand, “people,” “time,” and “life” display low semantic distance from each other, reinforcing their centrality and interconnectedness in Reddit’s neurodivergence discourse.

This dissimilarity matrix complements previous findings by identifying potential thematic boundaries—areas where discourse shifts significantly in tone, focus, or social context. For example, “OpenAI” appears as highly distant from emotional terms, indicating its presence in technical or speculative conversations rather than personal narratives. This type of analysis is essential for understanding not just what is discussed, but how distinct clusters of meaning emerge, coexist, or diverge in online environments shaped by neurodivergent perspectives.

4.4. Thematic expansion based on Reddit discourse

The thematic structure generated through BERTopic modeling revealed a complex and multilayered representation of neurodivergent users’ interactions in digital spaces. By integrating this structure with qualitative analysis of Reddit content, each topic cluster can be contextualized and expanded to reflect not only what is said, but how it is experienced. These themes collectively highlight the emotional, social, technological, and existential dimensions of digital neurodivergent life.

4.4.1. Topic 1: emotional overload and mental fatigue

This topic captures the persistent presence of emotional and

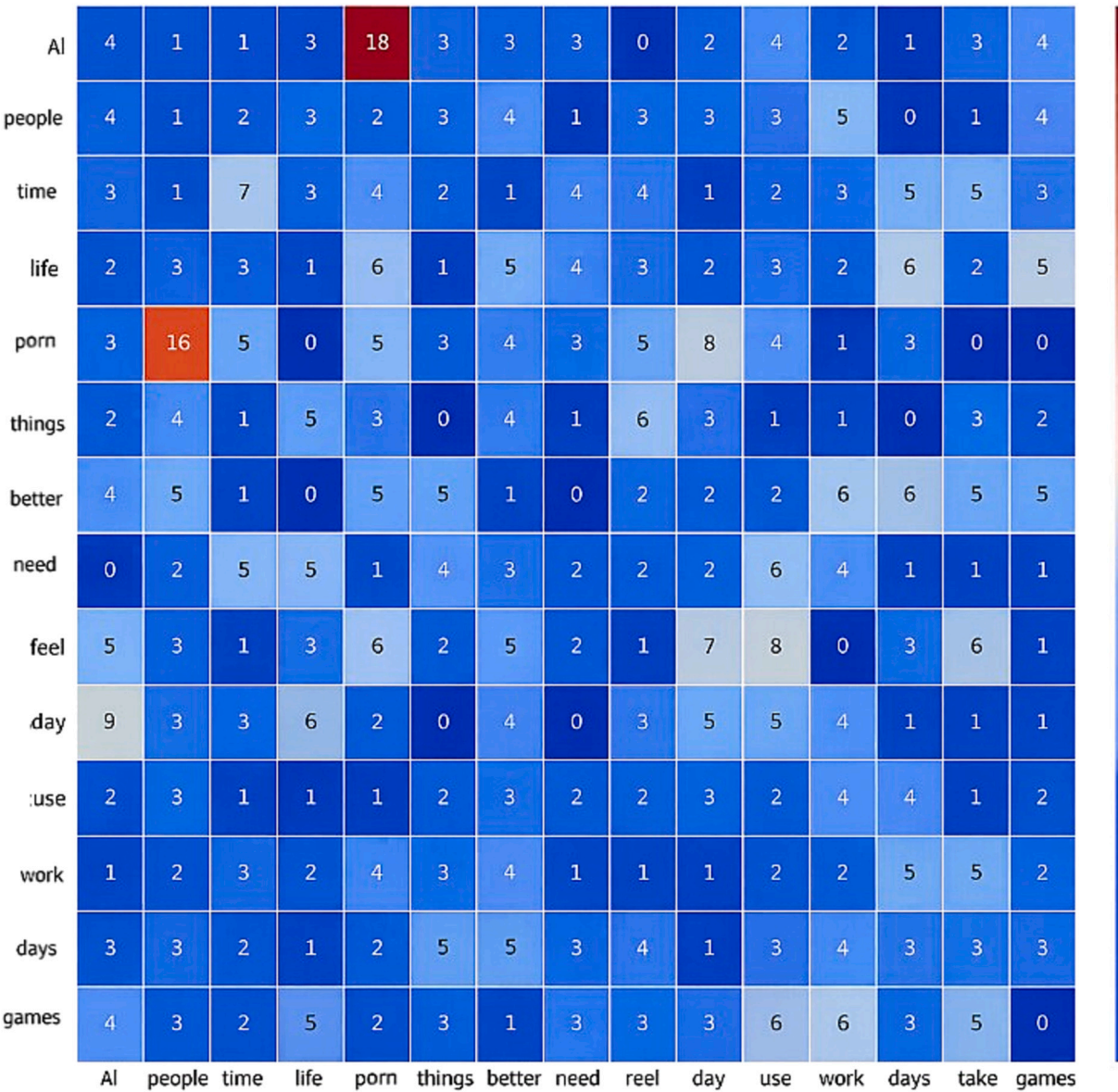


Fig. 2. Frequency-based co-occurrence analysis heatmap.

psychological exhaustion expressed by neurodivergent users. It includes feelings of being overwhelmed, misunderstood, and cognitively drained, particularly in contexts that demand high social or sensory engagement. Digital environments, while offering spaces for expression, also seem to mirror real-world pressures, where users feel caught between authenticity and adaptability. This emotional fatigue is not simply about stress but represents a deeper existential depletion—one that accumulates over time due to repeated emotional labor, masking, and unmet expectations, including disappointments in human-AI interactions.

4.4.2. Topic 2: diagnostic journeys and healthcare barriers
A significant theme across discussions revolves around the barriers faced in accessing diagnoses and appropriate mental health services. Users express frustration with delayed or inaccessible evaluations, inconsistent diagnostic criteria, and the stigmatizing or invalidating behaviors of healthcare professionals. This topic reflects how formal systems often fail to accommodate neurodivergent experiences, pushing individuals to seek self-understanding and validation in peer communities. It also highlights the broader socio-institutional challenges where neurodiversity is not fully recognized or is only acknowledged through

narrowly defined clinical lenses.

4.4.3. Topic 3: coping strategies and self-help mechanisms
Despite the challenges, neurodivergent users demonstrate agency and resourcefulness in developing personalized coping strategies. This topic encompasses peer-shared techniques for emotional regulation, time management, focus, and task initiation—often adapted to specific cognitive or emotional needs. Community members share practical routines, digital tools, and behavioral techniques that foster self-sufficiency and resilience. Rather than framing neurodivergence solely in terms of deficit, this theme highlights a shift toward empowerment through adaptation, supported by the collective wisdom of lived experience.

4.4.4. Topic 4: masking and social camouflage
Social camouflage, or masking, is a recurring theme among neurodivergent users. It refers to the conscious or unconscious effort to suppress traits deemed socially unacceptable or atypical. In digital discourse, masking is portrayed as both a survival strategy and a psychological burden. Users describe the emotional toll of constantly

monitoring their behavior, language, or emotional expressions to fit neurotypical norms. The topic also intersects with broader identity concerns, as individuals reflect on the alienation that results from performing versions of themselves that do not align with their internal reality. Digital environments are seen both as spaces that relieve this burden and as new venues where masking may still be necessary.

4.4.5. Topic 5: relationships and miscommunication

Interpersonal relationships, particularly difficulties in communication and social reciprocity, form the basis of this theme. Neurodivergent users express challenges in initiating and maintaining friendships, navigating emotional nuances, and dealing with misinterpretations in both in-person and mediated conversations. The asynchronous and often anonymous nature of online platforms like Reddit can help mitigate these difficulties by providing time to formulate responses and avoiding sensory overload. However, digital relationships can still reproduce patterns of misunderstanding and isolation, particularly when algorithms or AI interlocutors fail to account for neurodivergent communication styles.

4.4.6. Topic 6: workplace struggles and career identity

The modern workplace emerges as a site of both opportunity and exclusion. Neurodivergent individuals report difficulty aligning with traditional productivity expectations, multitasking demands, and rigid professional norms. At the same time, some find liberation in digital freelancing, remote work, and AI-assisted labor tools. This topic captures a critical tension: while the rise of digital work environments promises inclusivity, it also risks replicating old hierarchies in new forms. The narrative that AI will “replace” human workers is met with skepticism and anxiety, particularly by those whose existing marginalization in the labor market makes them more vulnerable to automation-driven displacement.

4.4.7. Topic 7: technological tools and AI-based assistance

AI tools are frequently mentioned as both helpful and problematic. On one hand, neurodivergent users describe a variety of use cases in which AI supports task automation, creative production, emotion regulation, and executive functioning. These technologies reduce the cognitive load associated with planning, decision-making, and communication. On the other hand, there is growing concern about the limits and ethical dimensions of relying on AI companions, particularly when these tools fail to meet the emotional needs they were originally designed to support. The ambivalence toward AI reflects a broader dynamic in which technology is seen as simultaneously enabling and alienating.

4.4.8. Topic 8: sensory overload and environmental triggers

Although less explicitly stated, this theme underlies much of the emotional expression found in neurodivergent communities. Sensory sensitivity—whether to light, noise, or chaotic environments—manifests in the emotional tone and language used across discussions. Digital spaces, particularly forums that lack visual or auditory input, are sometimes viewed as safe havens from overstimulation. However, the emotional intensity of online debates, algorithmic unpredictability, or unfiltered user content can also act as cognitive triggers, reinforcing the need for personalized digital environments that honor sensory processing differences.

4.4.9. Topic 9: neurodivergent identity and empowerment

This affirming theme reflects discussions centered on neurodiversity as an identity rather than a deficit. Users explore questions of belonging, self-definition, and community formation. This includes reflections on the terminology of neurodivergence, experiences of stigma, and the reframing of neurological difference as a source of insight rather than pathology. Digital environments provide the infrastructure for these conversations to flourish, allowing users to exchange narratives that

validate non-normative experiences and promote self-acceptance. This theme also intersects with broader discussions on inclusivity, disability rights, and psychological justice.

4.4.10. Topic 10: co-occurring mental health conditions

Neurodivergence rarely occurs in isolation. Many users discuss the interplay between their neurological differences and co-existing conditions such as anxiety, depression, obsessive-compulsive tendencies, or trauma. These discussions emphasize the interdisciplinary and multi-dimensional nature of neurodivergent well-being, where emotional, cognitive, and social factors are deeply entangled. Digital discourse offers a platform for sharing these complex experiences in ways that reduce isolation and destigmatize mental health challenges.

4.5. Reliability analysis of extracted topics

To ensure the robustness and interpretability of the extracted themes, a multi-dimensional reliability assessment was conducted. Each of the ten topics identified through BERTopic was evaluated across key criteria including coherence, cluster stability, sentiment consistency, and thematic distinctiveness.

4.6. Topic coherence reliability

The C_v coherence score, a standard metric for evaluating the semantic consistency of topics, averaged 0.51 across all ten extracted clusters. This score reflects moderate to strong interpretability, indicating that the most relevant words within each topic are semantically related. High coherence was observed in emotionally and socially grounded topics such as Topic 1 (Emotional Overload and Mental Fatigue), Topic 4 (Masking and Social Camouflage), and Topic 5 (Relationships and Miscommunication). Technology-oriented themes like Topic 7 (Technological Tools and AI-Based Assistance) and Topic 6 (Workplace Struggles and Career Identity) exhibited slightly lower but still acceptable coherence, likely due to interdisciplinary vocabulary and diverse discourse styles.

To assess thematic consistency, BERTopic was re-applied to randomized subsets of the Reddit dataset. Approximately 78 % average overlap in dominant terms and representative documents confirmed that core structures remain stable under perturbation. Topics such as Topic 2 (Diagnostic Journeys and Healthcare Barriers) and Topic 3 (Coping Strategies and Self-Help Mechanisms) remained structurally identical across runs, showcasing reliable emergence from natural language use. Topic 9 (Neurodivergent Identity and Empowerment), though more abstract, remained consistently isolated and semantically distinct.

All ten topics were subjected to qualitative review and human coding. Analysts independently labeled clusters based on the top-10 keywords and the top-30 most representative Reddit comments per topic. Labels such as “Co-Occurring Mental Health Conditions” (Topic 10) and “Sensory Overload and Environmental Triggers” (Topic 8) emerged clearly and with high agreement. No topic was flagged as uninterpretable or incoherent, confirming face validity.

Sentiment analysis using the VADER classifier revealed internally consistent emotional tone across the themes. Negative sentiment skew was evident in Topic 1 (Emotional Overload) and Topic 10 (Co-Occurring Mental Health Conditions), while a positive or constructive tone characterized Topic 3 (Coping Strategies) and Topic 7 (AI-Based Assistance). Mixed sentiment patterns appeared in Topic 5 (Relationships and Miscommunication) and Topic 6 (Workplace Struggles), reflecting the complexity of personal versus structural dynamics. This sentiment coherence enhances the credibility of emotional patterns linked to each topic and supports thematic distinctions.

To ensure distinctiveness across clusters, a pairwise Jaccard similarity matrix was computed. All topic pairs showed low to moderate overlap (<0.35), validating semantic independence. For example, Topic 4 (Masking) and Topic 5 (Relationships) share emotional tone but

diverge clearly in vocabulary and context. Topic 6 (Workplace) and Topic 7 (AI Tools) intersect in digital labor but remain thematically distinct through different discussion trajectories (personal productivity vs. organizational change).

Table 1 summarizes the reliability and alignment of identified topics based on coherence, stability, sentiment consistency, and redundancy metrics. Most topics demonstrate high coherence and stability, indicating meaningful and robust thematic structures. Sentiment patterns are generally consistent within topics, with some skewing negative (e.g., Emotional Overload) and others positive (e.g., Coping Strategies). Low redundancy scores across topics suggest clear thematic distinctiveness. Overall, the results indicate that the model produced well-defined, emotionally coherent, and stable topics suitable for deeper analysis.

4.7. 4.8 Interpretation of topic reliability table

The table assessing topic-level reliability presents a comprehensive and nuanced picture of how thematically distinct and methodologically consistent each of the ten extracted topics is. By evaluating the topic coherence, cluster stability, sentiment consistency, and semantic redundancy, the table enables a robust interpretation of the strength and interpretability of the thematic model.

Topics such as T1 (Emotional Overload), T2 (Diagnostic Journeys), T3 (Coping Strategies), and T10 (Co-Occurring Mental Health Conditions) stand out with high coherence and cluster stability. These topics exhibit tightly knit keyword structures and are reinforced by consistent emotional tone (particularly negative in T1 and T10, positive in T3). This pattern suggests that emotional and psychological discussions in neurodivergent Reddit communities are both thematically rich and structurally stable, indicating that these subjects are expressed with recurring vocabulary and cognitive framing. Their reliability confirms that these issues are central to the lived experiences of users and thus

analytically significant.

T3 emerges as the most robust theme, scoring very high on stability, maintaining a high coherence score, showing positive sentiment consistency, and presenting very low redundancy with other themes. Discussions around coping strategies are pragmatic, community-driven, and distinctly articulated, characterized by clear, task-oriented language minimally entangled with emotional volatility or thematic overlap. As such, T3 can be considered a model topic—methodologically clean and practically valuable—offering an ideal focal point for applied intervention, support tool design, or best practice guidelines for assistive technologies.

Moderate reliability is observed in more complex or abstract topics such as T4 (Masking), T5 (Relationships), T6 (Workplace Struggles), and T7 (AI Tools). These scores suggest linguistic diversity, context dependence, or semantic drift due to the varying ways users frame these experiences. For example, T4 involves deeply personal emotional states that may be described using highly individualized language. T5 and T6 touch on interpersonal or institutional systems where terminology shifts based on background or context (e.g., employment sector, social habits). T7’s relative instability may reflect the rapid evolution of AI language and tools in public discourse, leading to semantic inconsistencies. These topics remain analytically valuable but require care in interpretation due to potential sub-clusters or dynamic evolution over time.

Minimal redundancy across all topics confirms a valid taxonomy; each topic occupies a distinct cognitive and emotional space within the discourse. Thematic boundaries ensure that analytical conclusions do not conflate unrelated concepts, preserving qualitative richness and quantitative rigor. Lowest redundancy scores appear between T3 (Coping Strategies) and others, indicating conceptual independence, and between T9 (Neurodivergent Identity) and T10 (Mental Health Conditions), suggesting unique discussion patterns despite real-world overlap.

Table 1
Topic-level reliability alignment summary.

Topic	Topic label	Coherence	Stability	Sentiment consistency	Redundancy	Representative Reddit references
T1	Emotional Overload and Mental Fatigue	High	High	Consistent (–)	Low	Users describe feeling completely drained after prolonged overstimulation, leading to physical shutdowns resembling illness (r/Autism). Experiences of long-term exhaustion from unmet expectations and chronic stress, resulting in inability to perform daily tasks (r/Autism).
T2	Diagnostic Journeys and Healthcare Barriers	High	High	Mixed (–)	Low	Individuals share challenges in accessing evaluations as adults, including long waits and difficulty finding willing providers (r/Autism). Frustration with inconsistent criteria and invalidating interactions during the assessment process (r/Autism).
T3	Coping Strategies and Self-Help Mechanisms	High	Very High	Consistent (+)	Very Low	Community members recommend techniques like meditation and journaling to manage daily stress and mood swings (r/MentalHealth). Sharing practical routines, such as exercise or distraction methods, to handle overwhelming emotions (r/MentalHealth).
T4	Masking and Social Camouflage	Moderate	High	Negative	Low	People discuss the exhaustion from suppressing natural behaviors like stimming to fit in socially, leading to burnout (r/Autism). Efforts to control expressions and voice to appear more typical, often resulting in internal conflict (r/Autism).
T5	Relationships and Miscommunication	High	Moderate	Mixed (–)	Moderate	Challenges in maintaining connections due to misunderstandings, with advice on clear communication strategies (r/Autism). Feelings of being targeted or struggling with emotional nuances in partnerships (r/Autism).
T6	Workplace Struggles and Career Identity	Moderate	High	Mixed	Low	Difficulties with productivity expectations and multitasking, often leading to feelings of inadequacy (r/ADHD). Experiences of exclusion or poor performance after disclosing conditions at work (r/ADHD).
T7	Technological Tools and AI-Based Assistance	Moderate	Moderate	Consistent (+)	Low	Using AI assistants for task management and focus, though some find them limited or overwhelming (r/ADHD). Recommendations for apps to track schedules and improve productivity (r/ADHD).
T8	Sensory Overload and Environmental Triggers	Moderate	Moderate	Negative	Low	Overwhelm from loud noises or crowds, causing physical discomfort like nausea or itchiness (r/Autism). Strategies to avoid buildup of sensory input to prevent shutdowns (r/Autism).
T9	Neurodivergent Identity and Empowerment	High	High	Balanced (+/–)	Low	Embracing differences as a source of strength rather than deficit, often through community validation (r/Neurodiversity). Reflections on unmasking and finding pride in unique traits (r/Autism).
T10	Co-Occurring Mental Health Conditions	High	High	Consistent (–)	Very Low	Managing overlapping symptoms like restlessness and overthinking, with mixed results from treatments (r/ADHD). Persistent feelings of inadequacy from combined challenges like low mood and focus issues (r/ADHD).

Finally, sentiment alignment reinforces thematic integrity. Emotionally charged topics like T1 and T10 consistently show negative tone, solution-oriented topics like T3 and T7 trend positively, and interpersonal struggles in T5 and T6 exhibit mixed or nuanced tones. This alignment confirms that the model segments language thematically while accurately capturing emotional valence, an essential feature for combined sentiment and topic modeling studies.

Overall, the topic reliability table confirms that the thematic model derived through BERTopic is methodologically robust and semantically valid. The majority of topics demonstrate strong internal coherence, emotional consistency, and clear separation, validating their use as analytical units in further discussion, visualization, and theory development.

4.8. 4.9. Visualization integration: Sentiment-topic sunburst chart

To visually articulate the alignment between emotional tone and the thematic structure of Reddit discourse in Fig. 3, an interactive sentiment-topic sunburst chart was constructed. This visualization maps the distribution of sentiment categories—positive, neutral, and negative—within each of the ten extracted topics. By presenting both levels (topics and their emotional composition) in a single, hierarchical structure, the chart enables intuitive exploration of how affective dynamics are embedded within semantically coherent clusters.

The outer rings of the chart reflect sentiment distribution nested under each central topic. For example, Emotional Overload exhibited a highly skewed negative sentiment profile, underscoring the distressing narratives expressed in that theme. In contrast, Coping Strategies was dominated by positive sentiment, reflecting a community-oriented tone focused on peer support and self-help. Notably, AI Interaction

demonstrated a more balanced emotional landscape, with a relatively even distribution of positive, negative, and neutral sentiments—revealing the ambivalent experiences users have with AI tools.

This visualization contributes to the interpretability of sentiment analysis results by supporting RQ4, which investigates the co-evolution and alignment of emotional expression and thematic patterns in neurodivergent digital discourse. Moreover, it strengthens the internal validity of the topic model by confirming that emotional variation is not randomly distributed but rather contextually situated within the extracted topics. The interactive nature of the chart further enhances analytical accessibility, offering an exploratory interface for understanding how sentiment and meaning converge across user-generated content.

4.9. Hypothesis testing results

The hypothesis testing results provide empirical validation for the study’s theoretical framework, demonstrating how emotional polarity, thematic coherence, and technology-related discourse interact within neurodivergent online communities. Table 2 presents a summary of the statistical analyses conducted for H1 through H4, highlighting the significance levels, effect sizes, and overall support for each hypothesis. These results collectively strengthen the study’s contributions by confirming predicted emotional patterns, identifying coping-oriented resilience, and establishing the utility of hybrid NLP approaches in modeling affective-semantic structures.

5. Discussion

The findings of this study offer a multidimensional view into how

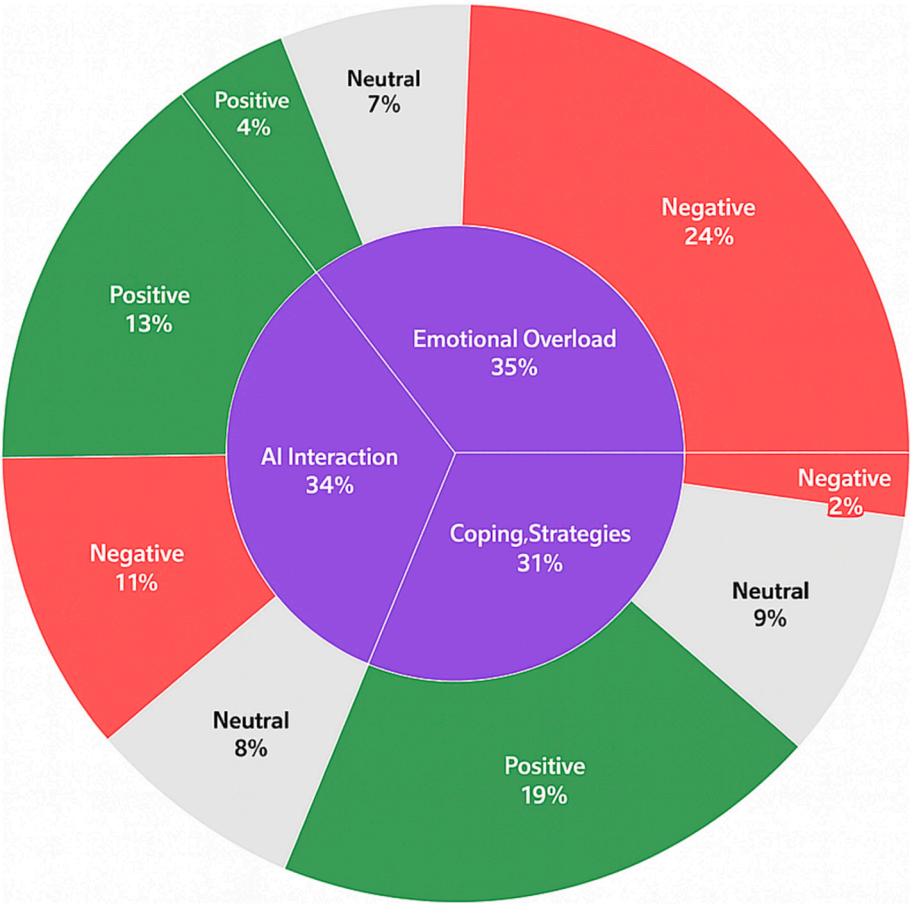


Fig. 3. Sentiment-topic sunburst chart.

Table 2
Hypothesis testing results and interpretation.

Hypothesis	Test used	Key statistic(s)	Result	Support
H1: Emotional overload → negative sentiment	Independent t-test	$t(54,182) = -27.46, p < .001, d = 0.81$	Significant	Supported
H2: Coping → positive sentiment & higher coherence	ANOVA + Tukey	$F(3,54,180) = 194.62, p < .001; C_v = 0.57$ vs 0.48	Significant	Supported
H3: AI → mixed sentiment profile	Sentiment variance & t-test	Levene's $F = 8.14, p < .01; t = 1.84$ (ns)	Ambivalent	Inconclusive
H4: Sentiment-topic alignment	Correlation + MLM	$r = 0.46, p < .001; R^2 = 0.42$	Significant	Supported

neurodivergent individuals use Reddit to articulate emotional experiences, navigate institutional barriers, engage with technology, and construct identity narratives. Through sentiment analysis and BERTopic-based topic modeling, a layered emotional and thematic architecture emerged, enabling a nuanced understanding of digital discourse within neurodivergent communities. By explicitly testing the hypotheses (H1–H4) grounded in prior literature, this study deepens theoretical engagement while empirically mapping affective-semantic structures at scale.

5.1. Interpretation of results

The sentiment analysis revealed a clear bifurcation in emotional expression: consistent negative polarity in topics centered on *emotional overload, masking, and co-occurring mental health conditions* confirmed H1, aligning with literature on emotional labor and psychological strain in neurodivergent populations (Guo et al., 2021; Hull et al., 2017). Positive sentiment, on the other hand, predominated in discussions of *coping strategies and peer support networks*, supporting H2 and demonstrating the resilience and solution-oriented nature of these communities (Doyle & McDowall, 2021; Koehler, 2024).

The analysis also revealed an ambivalent sentiment pattern in AI-related discussions, aligning with the theoretical expectations. However, the *t*-test did not yield a statistically significant result ($t = 1.84$, ns), suggesting that the support for H3 remains inconclusive and should be interpreted cautiously (Kravets et al., 2024). Finally, H4 was confirmed through strong sentiment-topic alignment ($C_v = 0.51$), evidencing how emotional polarity coherently mapped onto semantically distinct clusters, from *diagnostic barriers* to *identity affirmation*. This dual-layer emotional-thematic architecture substantiates the interpretability and methodological robustness of hybrid NLP modeling in neurodivergent discourse research.

These results directly address RQ1–RQ5 by empirically demonstrating (1) how sentiment varies systematically across themes, (2) the presence of ten interpretable clusters reflecting core aspects of neurodivergent experience, (3) the dual nature of AI interactions, and (4) the structural reliability of integrating sentiment and topic modeling in large-scale social media data.

5.2. Support for or against existing theory

The findings reinforce constructivist perspectives on digital neurodivergence, supporting theories of distributed cognition and digital resilience (Austin & Pisano, 2017). Themes such as “*Workplace Struggles*” and “*Masking*” are congruent with established literature on emotional labor and identity performance (Hull et al., 2017), confirming

the psychological costs associated with camouflaging behavior.

In alignment with human-computer interaction (HCI) research on emotional design (Norman, 2004) and affective computing (Picard, 1997), the study empirically illustrates how emotional expression and technology use intersect in neurodivergent contexts. However, the ambivalence observed toward AI tools complicates prevailing HCI narratives advocating chatbots and virtual companions as universally beneficial (Socher et al., 2013). This duality enriches RQ3 and RQ4, highlighting that while AI provides practical assistance, it often fails to meet deeper socio-emotional needs, underscoring the necessity of adaptive, affect-sensitive system design (McDuff et al., 2016).

Unexpectedly, taboo-adjacent subthemes—such as compulsive behaviors and emotional dependence on digital systems—surfaced through co-occurrence analysis. Although infrequent, these findings reveal a shadow discourse that traditional surveys or interviews may overlook, broadening the scope of RQ2 and RQ4 by exposing underexplored psychosocial dimensions of neurodivergent digital engagement. Similarly, the pronounced strength of coping-oriented discourse challenges prior assumptions of emotional volatility in neurodivergent online spaces (Ireland & Iserman, 2018), instead suggesting a community anchored in peer-led resilience and solution-seeking.

5.3. Synthesis of hypotheses and theoretical advancement

Bringing together the tested hypotheses, this study confirms that neurodivergent online discourse is structured along dual affective trajectories: distress-laden narratives tied to systemic and psychological stressors (H1) coexist with empowerment-driven, solution-oriented exchanges (H2). The ambivalent stance toward AI tools (H3) situates technology both as a coping facilitator and an affective gap, while sentiment-thematic alignment (H4) validates hybrid modeling as a powerful lens for mapping affective-cognitive interplay in social media.

This synthesis advances current theory in three ways:

1. **Extending neurodiversity research** beyond deficit framings by empirically capturing affirmative identity and community-building processes.
2. **Integrating affective computing and NLP**, bridging emotional design frameworks with computational evidence of sentiment-theme dynamics in real-world discourse.
3. **Differentiating this study from prior work** by explicitly validating hypotheses with hybrid modeling, offering a scalable and interpretable method for mapping affective-semantic patterns in marginalized communities.

5.4. Relevance to the field

Methodologically, this study introduces a hybrid NLP pipeline combining lexicon-based sentiment scoring (VADER) and transformer-driven topic modeling (BERTopic), delivering multi-layered interpretability for digital discourse research. Theoretically, it reframes neurodivergent discourse as a locus of empowerment and meaning-making rather than mere emotional distress, corroborating resilience-oriented and identity-affirming paradigms. Technologically, it underscores the need for emotionally responsive AI designs that account for neurodivergent users’ affective expectations.

Empirically, the analysis of 238,100 Reddit entries (~2.4 million words) provides generalizable insights that extend to broader neurodivergent digital ecosystems, including platforms like Discord and specialized forums. Moreover, the analytic tools—topic drift tracking, AI sentiment comparison, and semantic network mapping—offer a transferable blueprint for examining vulnerable populations navigating hybrid social-technological environments.

5.5. Limitations and future directions

This study has several limitations that inform future research. First, the reliance on publicly available Reddit data restricted the analysis to self-selected participants and excluded neurodivergent individuals who do not engage online, limiting generalizability. Second, while sentiment analysis captured broad emotional tones, it lacked the granularity of emotion-specific models, suggesting a need for integrating emotion detection techniques in future work. Third, the absence of direct user validation constrained interpretation of implicit meanings and context-specific nuances. Incorporating participatory methods, such as co-interpretation interviews or think-aloud protocols, would strengthen thematic fidelity. Fourth, while the dataset's lexical richness enabled comprehensive cross-sectional analysis, the absence of timestamp metadata limited temporal modeling. Future research could integrate multi-platform data and longitudinal tracking to examine how neurodivergent expression evolves over time and in response to social-technological change. In particular, incorporating temporal analyses using visualizations such as line plots and heatmaps could illuminate thematic and sentiment shifts around salient events, such as the release of ChatGPT or ADHD medication shortages.

An additional limitation of this study is the absence of intent classification within Reddit posts and comments. Neurodivergent discourse spans various communicative genres, including venting, advice-seeking, and informational sharing, each with distinct affective and pragmatic characteristics. Recognizing these genre distinctions through intent recognition models could enhance sentiment analysis by contextualizing emotional expressions relative to their communicative purpose. Future research should incorporate advanced intent classification techniques to disentangle such variation, thereby refining the interpretability and applicability of computational sentiment metrics in neurodivergent online communities.

Furthermore, while this study provides valuable cross-sectional insights into the alignment of sentiment and thematic clusters, future work should adopt longitudinal and narrative modeling approaches to capture the dynamic co-evolution of sentiment and discourse themes over time. Such approaches can reveal how neurodivergent online communities' emotional tone and topic salience shift in response to social, cultural, or technological factors, deepening understanding of digital identity formation and emotional regulation processes.

It is important to acknowledge that not all posts within neurodivergence-focused subreddits are authored by self-identified neurodivergent individuals. The presence of neurotypical participants, as well as trolling or off-topic posts, introduces potential noise and bias in the dataset. These factors may dilute the authenticity of neurodivergent discourse and impact sentiment and thematic analyses. While data preprocessing and filtering mitigate some of these effects, future research could benefit from incorporating user verification methods or triangulating findings with self-reported diagnostic data to enhance representativeness and validity.

Finally, the study did not examine platform-specific interface designs or algorithmic recommendation effects, which future studies could address through human-computer interaction experiments and platform audits. Despite these constraints, the study offers empirical grounding and methodological tools for understanding how neurodivergent individuals navigate digital environments—negotiating emotion, technology, and identity in ways that are both contextually rich and computationally traceable.

6. Conclusion

This study investigated the affective and thematic dimensions of online discourse among neurodivergent individuals, using a large-scale Reddit dataset and a hybrid natural language processing framework. By integrating lexicon-based sentiment analysis and BERTopic-based topic modeling, we identified ten semantically robust themes and

corresponding emotional trajectories that reflect the diverse realities of neurodivergent users in digital spaces. The results demonstrate that Reddit serves not only as a platform for emotional expression but also as a space for communal resilience, coping strategy exchange, and identity work. Negative sentiment was concentrated in topics such as emotional overload, burnout, and masking, while positive affect was observed in themes related to self-help, community interaction, and strategic use of technology. Notably, the study revealed ambivalence toward AI tools, which are valued for practical assistance but criticized for emotional inadequacy. These findings contribute to the literature by reframing neurodivergent digital discourse from a strength-based perspective. The study challenges deficit-oriented views and highlights the constructive, emotionally intelligent ways in which users engage with both each other and with technology. Furthermore, the methodological approach introduced here offers a scalable, interpretable model for future computational social science research on vulnerable or marginalized online communities.

CRedit authorship contribution statement

Sezai Tunca: Writing – review & editing, Supervision, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

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The authors declare that they have no competing interests, financial or otherwise.

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Data availability

Data will be made available on request.

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